26F Stat TA Session W5

📢 Quiz 2 Information

- Time: **10/15 (Thu.) 15:30 - 16:20** (first hour of TA session)
- **We will have TA Session after the quiz!**
- Coverage: Chapter 3 & 4 (no TA session material included)
- Calculators are permitted
- **Reminder:** `R` quiz will be held at Week 7. Review the TA session materials!

1. More Distributions

- Recall that we learn how to generate random numbers with `runif()` (uniform distribution)
- As the lecture continues, we'll be familiar with more common distributions. e.g., **binomial distribution** and **normal distribution**

Binomial Distribution

- Before we are introduced to Binomial distribution, we first recall the concept of **Bernoulli distribution**
- A random variable following Bernoulli distribution assigns 1 if the trial succeeds; 0 if the trial fails

$$X \sim \text{Bernoulli}(p),\ X = \begin{cases} 1, & \text{if the trial succeeds} \\ 0, & \text{if the trial fails} \end{cases}$$

- **Binomial distribution** is a close related to the Bernoulli distribution. It counts **the number of successful trials** among a total n independent trails
- Now let's draw random samples from binomial distribution using `rbinom()`

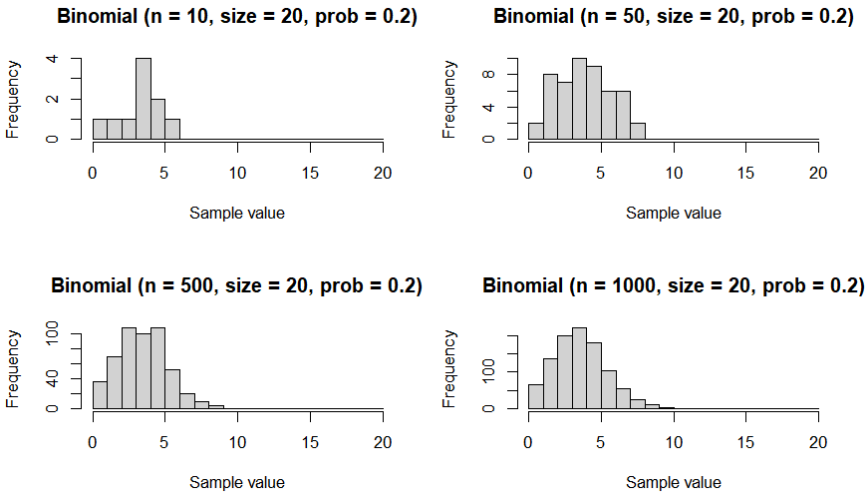


Figure: Random Samples drawn from Binomial(20, 0.2) in different sizes

```r
rdnum <- rbinom(n = 30, size = 20, prob = 0.5)
# n = number of observations
# size = number of trials
```

Normal Distribution

- Normal distribution models data that are centered around the mean, with most values close to the mean and fewer values farther away

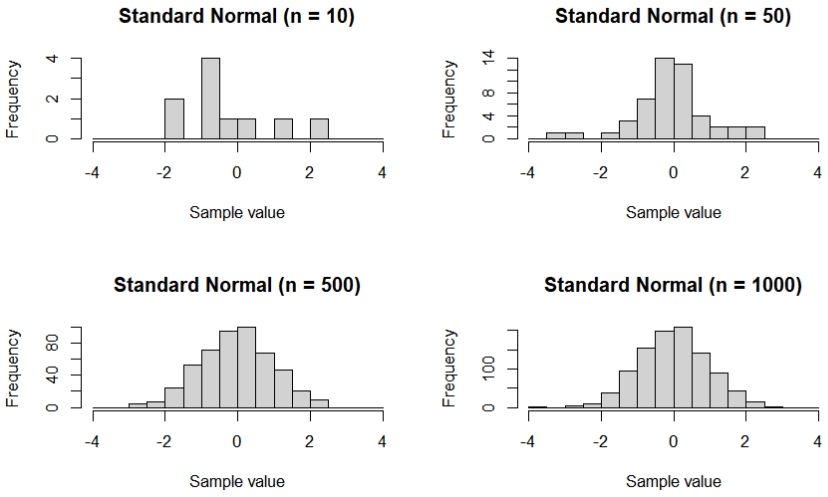


Figure: Random Samples drawn from N(0,1) in different sizes

```r
rdnum <- rnorm(10) # default: mean = 0, sd = 1
```

```r
rdnum_2 <- rnorm(n = 10, mean = 60, sd = 15)
```

2. Control Flow

- By default, a program executes instructions one after another. **Control flow** allows the program to change this order by repeating actions with `for` loop or by making decisions with `if-else` statements

`for` loop

- A for loop repeats the same task over and over for each item in a list or for a fixed number of times

```r
for (i in 1:5){
    print(i)
    }

# output
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
```

`if-else`

- An `if-else` statement tells the computer what to do based on whether a condition is `TRUE` or `FALSE`

```r
# syntax
if(a < b){
    # do work 1
    }else if(a == b){
        # do work 2
        }else{
            # do work 3
            }

# example
for (i in 55:65){
  if(i < 60){
    print("Fail:(")
    }else if(i == 60){
      print("On the threshold!")
      }else{
        print("Pass:)")
```

```
  }
}
```

`replicate()`

- The `replicate()` function is another way to repeat an expression multiple times. It stores the results in a vector, matrix, or list

```
rep_value <- replicate(n = 3, "Hello!")
rep_sample <- replicate(n = 4, rnorm(2, mean = 0, sd = 1))

# output
[1] "Hello!" "Hello!" "Hello!"

           [,1]        [,2]       [,3]      [,4]
[1,] -0.102854  1.44482927  0.5796968 1.2482826
[2,]  1.159166 -0.03620607 -0.7763544 0.7197338

# the output of a single trial is stored in columns
```

`while()`

- `while()` allows us to keep running the same code until a condition is satisfied

```
i <- 1

while (i <= 10) {
  print(i)
  i <- i + 1
}

# what is the output?
```

A Dice Rolling Example

- Recall that we learned how to simulate a coin toss situation using `R` last week!
- Now we want to simulate the a dice roller
  - We can use `sample()` to randomly select a number (similar to `runif()` in coin flip)

```
# case 1: did i get a six?

set.seed(123)
sim <- sample(1:6, 1)  # 1:6 represents the six outcomes; 1 means we roll one
time
```

```r
if (sim == 6) {
    print("we got a six!")
}   else {
    print("nice try")
}

# output
[1] "nice try"

# case 2: how many six we get in 10 rolls

for (i in 1:10){
    sim <- sample(1:6, 1)
    if (sim == 6) {
        print(1)
    }   else {
        print(0)
    }
}

# case 3: keep rolling until we make it

sim <- sample(1:6, 1)

while (sim != 6) {          # != means unequal
  print(sim)
  sim <- sample(1:6, 1)
}

print(sim)

# when will we need to use replicate()?
```

3. Define Your Functions!

- Besides learning how to use functions in packages, we can also define our own functions with `function()`

```r
# syntax
function_name <- function(param1, param2, ...){
    # do something
    return(output)
}
```

```r
# example
add <- function(a, b){
    return(a+b)
}

add(50, 100) # 150
```

- There are several reasons to define your own functions
  - Avoid copy-paste the same code for repeated use
  - Customize the parameters
  - You can save functions in a `R` script (e.g., `myfunctions.R`) and import it when you need it!

4. Assign Column Names

```r
df <- data.frame(a = 1:3, b = 4:6)

# output
  a b
1 1 4
2 2 5
3 3 6

colnames(df) <- c("height", "weight")

# output
  height weight
1      1      4
2      2      5
3      3      6
```

Exercises

W5 Exercises